

Learner Demographics and Course Enrollment Behavior Analysis on EduPro

Research Paper

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Abstract

Online learning platforms generate large volumes of learner interaction data. Understanding learner demographics and enrollment behavior helps educational platforms improve engagement, personalize learning experiences, and optimize course offerings. This research focuses on descriptive analytics of the EduPro platform by examining age groups, gender participation, course preferences, and learner behavior patterns. Using Python, Pandas, Plotly, and Streamlit, an interactive analytics dashboard was developed to visualize enrollment trends and learner intelligence insights.

1. Introduction

Digital education platforms have transformed modern learning by providing flexible and accessible educational resources. Platforms such as EduPro serve learners from diverse demographic backgrounds with different educational goals and interests.

Understanding:

- * who the learners are,
- * which courses they prefer,
- * how enrollment varies across demographics,

is essential for designing better learning experiences and improving platform engagement.

This project performs descriptive learner intelligence analysis rather than predictive modeling. The study aims to provide meaningful insights into learner demographics and course enrollment behavior.

2. Problem Statement

EduPro currently stores learner and transaction data but lacks a centralized analytical system to answer important business questions such as:

- * Which age groups are most active on the platform?
- * Which course categories attract the highest enrollments?
- * Are there gender-based differences in course preferences?
- * Which skill levels are most preferred?
- * How do learner demographics influence enrollment behavior?

Without these insights, strategic decisions related to course development, learner engagement, and marketing remain intuition-driven rather than data-driven.

3. Objectives

The primary objectives of this project are:

1. Analyze learner demographic distribution.
2. Study course enrollment behavior.
3. Identify course popularity patterns.
4. Compare enrollment trends across age groups and genders.
5. Build an interactive Streamlit dashboard for visualization.
6. Generate actionable learner intelligence insights.

4. Dataset Description

The dataset consists of three interconnected sheets:

4.1 Users Sheet

Contains learner demographic details.

Fields:

- * UserID

- * AgeGroup

- * Gender

4.2 Courses Sheet

Contains course-related information.

Fields:

- * CourseID

- * CourseName

- * CourseCategory

- * CourseType

- * CourseLevel

4.3 Transactions Sheet

Contains learner enrollment transactions.

Fields:

- * TransactionID

- * UserID

- * CourseID

- * TransactionDate

5. Methodology

The project follows a step-by-step analytical methodology.

5.1 Data Integration

The datasets were merged using:

- * UserID
- * CourseID

This created a unified master dataset for analysis.

5.2 Data Cleaning

The following preprocessing steps were performed:

- * Handling missing values
- * Removing duplicate entries
- * Formatting date columns
- * Standardizing category labels

5.3 Exploratory Data Analysis (EDA)

The analysis included:

- * Age distribution analysis
- * Gender participation analysis
- * Course category popularity
- * Course level distribution
- * Demographic vs course preference analysis

5.4 Visualization

Interactive visualizations were developed using:

- * Plotly Express

- * Streamlit

Charts included:

- * Bar charts

- * Heatmaps

- * Sunburst charts

- * Enrollment trend charts

6. Tools and Technologies Used

Technology	Purpose	
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Python	Core programming	
Pandas	Data manipulation	
NumPy	Numerical operations	
Plotly	Interactive visualizations	
Streamlit	Dashboard development	
VS Code	Development environment	

7. Key Analytical Questions

This project aims to answer:

- * What is the age distribution of learners?

- * Which course categories are most popular?

- * Which demographic groups enroll the most?

- * Do gender-based differences exist in course selection?

- * Are beginner or advanced courses more preferred?

8. Results and Findings

8.1 Age Distribution

The majority of learners belong to:

- * 18–25 age group

- * 26–35 age group

These groups represent the highest engagement levels on the platform.

8.2 Gender Participation

The platform demonstrates relatively balanced gender participation with slightly higher engagement from female learners in selected categories.

8.3 Course Popularity

Most enrolled categories include:

- * Development

- * Data Science

- * Business

Technical and skill-oriented courses showed strong enrollment concentration.

8.4 Skill-Level Preferences

Beginner-level courses recorded the highest enrollments, indicating strong participation from early-stage learners.

Advanced courses had comparatively lower enrollment counts due to higher learning complexity.

8.5 Behavioral Insights

Key behavioral observations include:

- * Younger learners prefer technical skill courses.
- * Advanced learners enroll in specialized content.
- * Enrollment concentration is higher among active repeat learners.

9. Dashboard Features

The Streamlit dashboard includes:

- * Interactive sidebar filters
- * KPI monitoring
- * Demographic analysis
- * Enrollment visualization
- * Heatmaps
- * Behavioral analysis
- * High-dimensional data exploration

10. Key Performance Indicators (KPIs)

The dashboard tracks:

- * Total enrollments
- * Unique active learners
- * Gender participation ratio

- * Course popularity index

- * Skill-level distribution

11. Advantages of the System

The proposed analytics dashboard provides:

- * Better learner understanding

- * Improved course planning

- * Data-driven decision making

- * Enhanced learner engagement

- * Simplified educational intelligence reporting

12. Limitations

Current limitations include:

- * Use of descriptive analysis only

- * No predictive recommendation engine

- * Synthetic/mock dataset generation

- * Limited real-world personalization

13. Future Enhancements

Future improvements may include:

- * Machine learning-based recommendations

- * Real-time analytics

- * Cloud deployment

- * Personalized learning paths
- * Advanced predictive modeling
- * User authentication systems

#14. Conclusion

This project successfully demonstrates how learner demographic and enrollment data can be transformed into actionable educational intelligence using Python and Streamlit.

The EduPro analytics dashboard provides a clear understanding of:

- * learner demographics,
- * course preferences,
- * enrollment behavior,
- * and engagement trends.

By leveraging data visualization and interactive analytics, EduPro can make informed decisions regarding course development, marketing strategy, and learner engagement optimization.